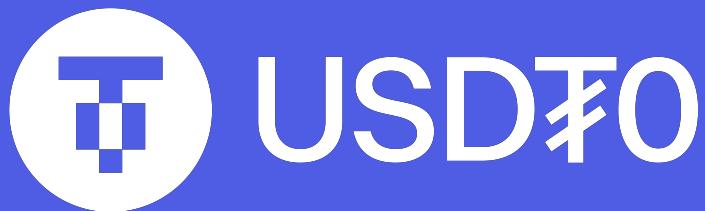


Everdawn USDT0 IOTA Deployment Assessment



March 24, 2026

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Summary

Type	Token	Total Issues	1 (0 resolved)
Timeline	From 2026-03-23 To 2026-03-23	Critical Severity Issues	0 (0 resolved)
Languages	Move	High Severity Issues	0 (0 resolved)
		Medium Severity Issues	0 (0 resolved)
		Low Severity Issues	0 (0 resolved)
		Notes & Additional Information	1 (0 resolved)
		Client Reported Issues	0 (0 resolved)

Scope

This report provides a detailed assessment of the USDT0 deployment on IOTA L1 mainnet, including package bytecode verification, deployment transaction review, and LayerZero pathway wiring verification, and privileged capability ownership-transfer verification.

The assessment was performed using the implementation addresses provided within the [Everdawn-Labs/usdt0-oft-contracts](https://github.com/Everdawn-Labs/usdt0-oft-contracts) repository and the OFT and Token packages for IOTA introduced in [PR#198](#) within the same repository.

System Overview

USDT0 on IOTA is deployed as a Move-based OFT/Oapp integration that connects to 22 LayerZero pathways. The IOTA side contains the authoritative peer table and pathway configuration state for outbound routing (send/receive library, ULN config, executor config, and enforced options).

This assessment focused on confirming:

1. Deployed package bytecode matches expected repository outputs.
2. The complete deployer transaction sequence is consistent and successful.
3. Current IOTA on-chain wiring state matches intended pathway mapping.
4. Current IOTA on-chain wiring state matches the intended configurations as specified in repository.
5. Privileged token/package/OFT capabilities were transferred and custodied as stated by the client.

Approach

The deployment assessment process consisted of the following checks:

1. Bytecode equivalence checks for deployed packages.
2. Deployment transaction integrity and initialization correctness checks.
3. LayerZero pathway configuration and peer-mapping consistency checks on IOTA.
4. Privileged capability ownership and custody verification.
5. Bidirectional readiness status check for remote reciprocal peering.

Deployment Verification

Core Deployment Identifiers

Role	Address
Deployer	0x4f1610...2a04
IOTA OFT Object	0x7fdb96...addd
IOTA OApp Object	0x6bd5f8...3216
IOTA OFT Package ID	0xe6a11e...5902
IOTA Token Package ID	0x25afea...7ac9
IOTA EID	30423
Peer Table Object	0xf4bc48...f2d1
Coin Metadata	0x1289f8...6d3b
Messaging Channel	0x8df497...62e6

Bytecode Verification

Package	Package ID	Result
Token (token)	0x25afea...7ac9	Matched expected bytecode (1/1 modules)
OFT (oft)	0xe6a11e...5902	Matched expected bytecode (14/14 modules)

Deployment Transaction Summary

All transactions from the deployer were retrieved and reviewed.

Operation	Count	Notes
Publish	2	Package deployments

Operation	Count	Notes
<code>oft_impl::init_oft</code>	1	
<code>oft_ptb_builder::lz_receive_info</code>	1	
<code>oft::register_oapp</code>	1	
<code>bytes32::from_bytes</code>	22	Helper for <code>set_peer</code>
<code>oapp::set_peer</code>	22	22 remote chains
<code>endpoint_calls::set_send_library</code>	22	All use same ULN302
<code>endpoint_calls::set_receive_library</code>	22	All use same ULN302
<code>endpoint_calls::set_config</code> + <code>uln_302::set_config</code>	66	22 send ULN + 22 receive ULN + 22 executor
<code>oapp::set_enforced_options</code>	44	22 chains x 2 message types
<code>TransferObjects</code>	7	Init + final cleanup
Total Transactions	161	
Successful Transactions	161	
Total MoveCall Operations	267	

Capability and Ownership Transfer Verification

The client stated that ownership was transferred to multisig address `0xfb11bd...9174`. The following privileged objects were reviewed in deployment and transfer transactions:

- OFT package `UpgradeCap 0xa45eda...a191` was transferred in `BZk8dC...0xZ15` to multisig `0xfb11bd...9174`.
- OApp `AdminCap 0xe277b4...01c8` was transferred in `BZk8dC...0xZ15` to multisig `0xfb11bd...9174`.
- Token package `UpgradeCap 0xb8adc8...2150` was transferred in `3VtKeo...eGb9J` to multisig `0xfb11bd...9174`.
- Token `DenyCapV1<USDT0> 0x8616da...e922` was transferred in `3VtKeo...eGb9J` to multisig `0xfb11bd...9174`.

- **MigrationCap** `0x0f8192...733c` was created in `n6NLVi...puYGS` and transferred in `7oWEyw...brVev` to multisig `0xfb11bd...9174`.
- Treasury capability input to `init_offt` `0xd1811c...5f9f` was wrapped in OFT `0x7fdb96...addd` during `n6NLVi...puYGS`.

All items above were confirmed from on-chain transaction/object state and match the client-provided ownership transfer statements.

IOTA Pathway Configuration State (Current)

Current IOTA wiring state was reconstructed from on-chain events and cross-checked against intended pathway mappings.

EID	Chain	Peer	Receive Confirmations	MsgType1 Gas	MsgType2 Gas
30101	Ethereum	<code>0x6c96de...1dee</code>	15	80000	100000
30109	Polygon	<code>0x6ba103...9e13</code>	32	80000	100000
30110	Arbitrum	<code>0x14e4a1...8d92</code>	60	80000	100000
30111	Optimism	<code>0xf03b4d...a0ad</code>	450	80000	100000
30181	Mantle	<code>0xcb768e...59cc</code>	2000	80000	100000
30212	Conflux	<code>0xc57efa...d84a</code>	2800	150000	150000
30274	XLayer	<code>0x94bcca...5554</code>	9000	80000	100000
30280	Sei	<code>0x56fe74...cc60</code>	2000	150000	150000
30295	Flare	<code>0x567287...7588</code>	500	80000	100000
30316	Hedera	<code>0xe3119e...53d6</code>	60	80000	100000
30320	Unichain	<code>0xc07be8...3ec3</code>	450	80000	100000
30322	Morph	<code>0xcb768e...59cc</code>	1000	80000	100000
30331	Corn	<code>0x3f8294...34e4</code>	2800	80000	100000
30333	Rootstock	<code>0x1a594d...6b61</code>	120	80000	100000
30339	Ink	<code>0x1cb6de...4a65</code>	450	80000	100000

EID	Chain	Peer	Receive Confirmations	MsgType1 Gas	MsgType2 Gas
30362	Bera	0x3dc963...79cb	60	80000	100000
30367	HyperEVM	0x904861...7e98	43200	80000	100000
30383	Plasma	0x02ca37...7cf9	1800	80000	100000
30390	Monad	0x915143...ecc5	3600	240000	240000
30396	Stable	0xedaba0...b027	3600	80000	100000
30398	MegaETH	0x915143...ecc5	5400	200000	200000
30410	Tempo	0xaf37e8...47ff	3000	550000	550000

Global config invariants on IOTA side:

- Send library (all pathways):
0x042e3bb837e5528e495124542495b9df5016acd011d89838ae529db5a814499e
- Receive library (all pathways):
0x042e3bb837e5528e495124542495b9df5016acd011d89838ae529db5a814499e
- Executor (all pathways):
0x29b691f9496eea6df8f4d77ceacee5949e92e7e51b2e3c2e6cd70eef5237e99a
- Max message size: 10000
- Send confirmations: 1500 across all 22 pathways (matches client-provided outbound confirmation parameter)
- Required DVNs (send/receive):
- 0x1f4fa24418593ee8087cb62203c7405e7cb7234edc42494aeface57c1f42eeac (USDT0 DVN)
- 0xa560697328ccb5dc3f3f8e8a2c41e282827060da7a29971d933e9aa405c2ba7f (LayerZero Labs DVN)

On-chain state matched the intended configuration for all receive confirmations, enforced options, and peer mappings

Remote-Side Bidirectional Readiness (Status)

A functional smoke-test for bidirectional readiness was attempted, but could not be completed because reciprocal remote-side peer configuration is not yet set on reachable chains.

This does not invalidate IOTA-side deployment correctness, but it indicates bidirectional pathway readiness remains incomplete.

Trust Assumptions and Privileged Roles

The following trust assumptions were identified during this deployment assessment:

- The LayerZero endpoint and messaging libraries on IOTA and remote chains are trusted to operate as specified.
- OApp/OFT privileged configuration operations (peer wiring, library selection, ULN config, executor config, options) are assumed to be controlled by authorized operators.
- Package/token/admin/migration privileged capabilities were observed transferred to multisig address [0xfb11bd...9174](#); post-deployment control and upgrade risk therefore depend on that multisig governance and signer security.
- Cross-chain operational readiness depends on both sides of each pathway being configured. This assessment confirms IOTA-side state; remote-side reciprocal peering is still pending.

Notes & Additional Information

N-01 MegaETH Peer Configuration does not Match with LayerZero's Recommendation

[LayerZero documentation](#) suggests the following:

MegaETH uses a multidimensional gas model (<https://docs.megaeth.com/megaevm#multidimensional-gas-model>), there are two gas costs associated with any given transaction - compute gas and storage gas. What this means for developers is that you should estimate the gas costs for MegaETH transactions beforehand and make sure you set enforcedOptions with the estimated amount + some buffer. As per our calculations - 250k gas should be sufficient.

In IOTA configurations, peer MegaETH has `200_000` gas limit for both message types. While this does not necessarily means the transactions would fail, it does not matches with what LayerZero suggests should be sufficient.

Consider updating the gas limit for MegaETH to `250_000` to be consistent with LayerZero, or if it is verified that `200_000` is sufficient, consider documenting it.

Note: Review of configuration choices relative LayerZero's recommendations was out of scope for this deployment review. This issue is shared for visibility.

Conclusion

OpenZeppelin reviewed the IOTA deployment artifacts for USDT0, including package bytecode evidence, full deployer transaction history, current LayerZero wiring state, and privileged capability ownership handoff.

The IOTA-side deployment and on-chain configuration are consistent with the intended design. No issues were identified within the scope of the deployment review. A minor concern related to configuration choices, outside the scope of this review, was included as an informational note.

As of time of assessment, reciprocal remote-side peering to IOTA EID [30423](#) was not yet configured on reachable EVM chains. As such, the attempted bidirectional smoke-test should be considered pending this prerequisite.

The Everdawn team is appreciated for clear communication and collaboration during the review process.

Appendix

Issue Classification

OpenZeppelin classifies smart contract vulnerabilities on a 5-level scale:

- Critical
- High
- Medium
- Low
- Note/Information

Critical Severity

This classification is applied when the issue's impact is catastrophic, threatening extensive damage to the client's reputation and/or causing severe financial loss to the client or users. The likelihood of exploitation can be high, warranting a swift response. Critical issues typically involve significant risks such as the permanent loss or locking of a large volume of users' sensitive assets or the failure of core system functionalities without viable mitigations. These issues demand immediate attention due to their potential to compromise system integrity or user trust significantly.

High Severity

These issues are characterized by the potential to substantially impact the client's reputation and/or result in considerable financial losses. The likelihood of exploitation is significant, warranting a swift response. Such issues might include temporary loss or locking of a significant number of users' sensitive assets or disruptions to critical system functionalities, albeit with potential, yet limited, mitigations available. The emphasis is on the significant but not always catastrophic effects on system operation or asset security, necessitating prompt and effective remediation.

Medium Severity

Issues classified as being of medium severity can lead to a noticeable negative impact on the client's reputation and/or moderate financial losses. Such issues, if left unattended, have a moderate likelihood of being exploited or may cause unwanted side effects in the system.

These issues are typically confined to a smaller subset of users' sensitive assets or might involve deviations from the specified system design that, while not directly financial in nature, compromise system integrity or user experience. The focus here is on issues that pose a real but contained risk, warranting timely attention to prevent escalation.

Low Severity

Low-severity issues are those that have a low impact on the client's operations and/or reputation. These issues may represent minor risks or inefficiencies to the client's specific business model. They are identified as areas for improvement that, while not urgent, could enhance the security and quality of the codebase if addressed.

Notes & Additional Information Severity

This category is reserved for issues that, despite having a minimal impact, are still important to resolve. Addressing these issues contributes to the overall security posture and code quality improvement but does not require immediate action. It reflects a commitment to maintaining high standards and continuous improvement, even in areas that do not pose immediate risks.